MPU Power & Clock

24MHz_IN — 24MHz_IN
$\overline{\text{RESET}}$ — $\overline{\text{RESET}}$

File: mpu_power_clock.kicad_sch

MPU Memory

$\overline{\text{RESET}}$ — $\overline{\text{RESET}}$

File: mpu_memory.kicad_sch

MPU Interface

$\overline{\text{RESET}}$ — $\overline{\text{RESET}}$

File: mpu_interface.kicad_sch

MPU Extra Pins

24MHz_IN — 24MHz_IN

File: mpu_dsp_fpga.kicad_sch

Changes from Rev A
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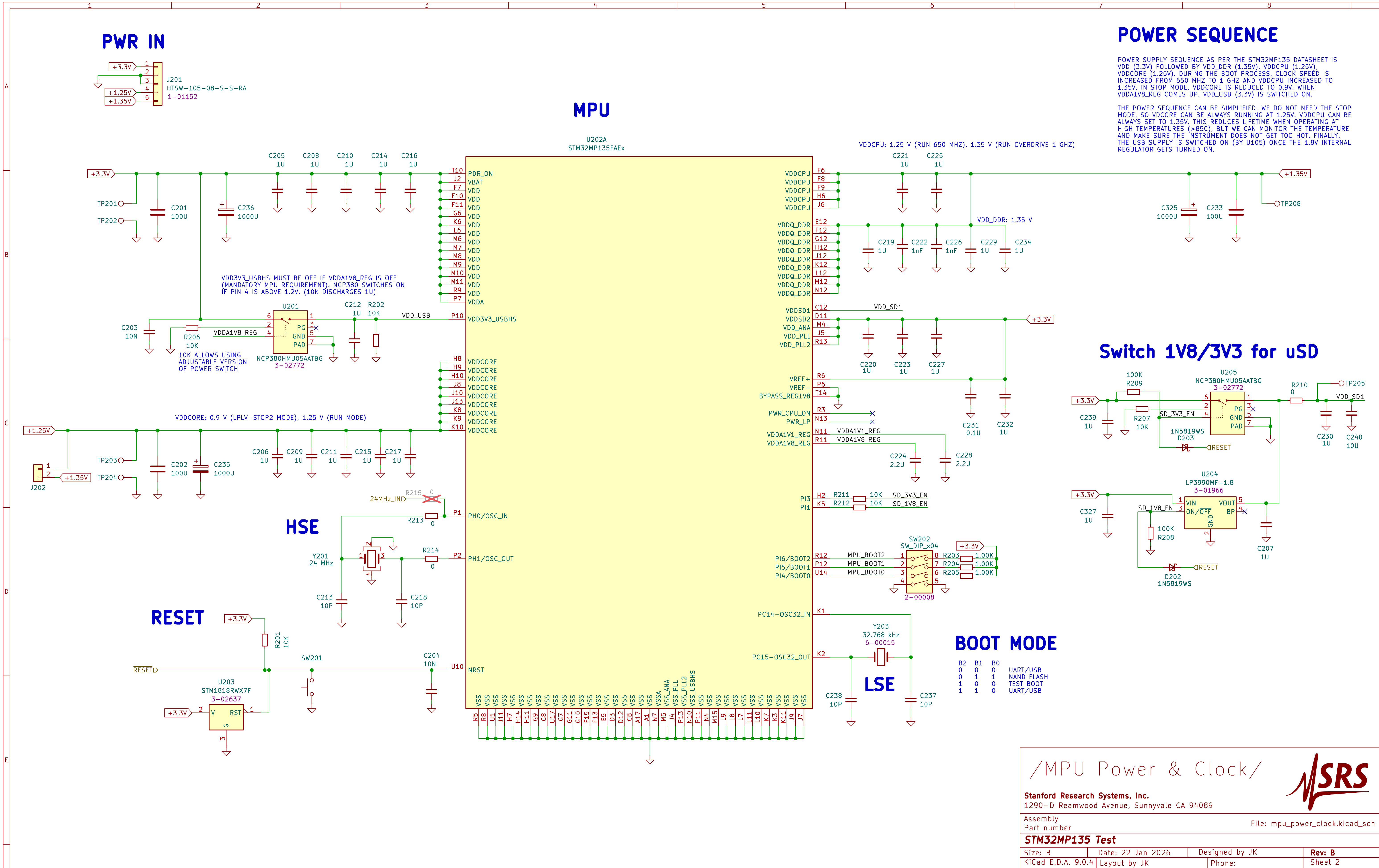
- Open solder mask over the JTAG connector
- Change NCP380 to fixed-current model and add current-limit resistor
- Remove the 1.5K pullup on USB D+ line
- Add 10k pulldown to VDD_NAND
- Tie Ethernet magnetics center taps to 3.3V via the choke
- Tie LED2 anode (pin 9, +LEDR) to VDD2A (3.3V) instead of GND
- Add more large capacitors to all power rails
- Add power input header and jumper to short 1.25V and 1.35V
- Reverse all LCD and CTP pins
- Swap NAND ALE and CLE wires, tie pin 48 to GND
- Add 0.1U to NAND_WP#
- Add green debug LED
- Add LSE crystal (32.768 kHz)
- Add 100k pulldown for CTP RST
- Label connector pins
- Add reset supervisor instead of 1U
- VDDSD1 switch 1V8/3V3, and NRST turns uSD power off
- Swap upper/lower DDR3L mask & strobe
- Do not swizzle DQ0 and DQ8 on DDR3L
- Add RS-232 receiver (MAX3232) and DB9 connector
- External 24 MHz input option

Stanford Research Systems, Inc.
1290-D Reamwood Avenue, Sunnyvale CA 94089

Assembly Part number		File: mb.kicad_sch	
STM32MP135 Test			
Size: B	Date: 22 Jan 2026	Designed by JK	**Rev: B**
KiCad E.D.A. 9.0.4	Layout by JK	Phone:	Sheet 1



PWR IN
J201
HTSW-105-08-S-S-RA
1-01152
MPU
U202A
STM32MP135FAEx
POWER SEQUENCE
POWER SUPPLY SEQUENCE AS PER THE STM32MP135 DATASHEET IS VDD (3.3V) FOLLOWED BY VDD_DDR (1.35V), VDDCPU (1.25V), VDDCORE (1.25V). DURING THE BOOT PROCESS, CLOCK SPEED IS INCREASED FROM 650 MHZ TO 1 GHZ AND VDDCPU INCREASED TO 1.35V. IN STOP MODE, VDDCORE IS REDUCED TO 0.9V. WHEN VDDA1V8_REG COMES UP, VDD_USB (3.3V) IS SWITCHED ON.
THE POWER SEQUENCE CAN BE SIMPLIFIED. WE DO NOT NEED THE STOP MODE, SO VDCORE CAN BE ALWAYS RUNNING AT 1.25V. VDDCPU CAN BE ALWAYS SET TO 1.35V. THIS REDUCES LIFETIME WHEN OPERATING AT HIGH TEMPERATURES (>85C), BUT WE CAN MONITOR THE TEMPERATURE AND MAKE SURE THE INSTRUMENT DOES NOT GET TOO HOT. FINALLY, THE USB SUPPLY IS SWITCHED ON (BY U105) ONCE THE 1.8V INTERNAL REGULATOR GETS TURNED ON.
VDDCPU: 1.25 V (RUN 650 MHZ), 1.35 V (RUN OVERDRIVE 1 GHZ)
VDD_DDR: 1.35 V
VDD3V3_USBHS MUST BE OFF IF VDDA1V8_REG IS OFF (MANDATORY MPU REQUIREMENT). NCP380 SWITCHES ON IF PIN 4 IS ABOVE 1.2V. (10K DISCHARGES 1U)
10K ALLOWS USING ADJUSTABLE VERSION OF POWER SWITCH
NCP380HMU05AATBG
3-02772
VDDCORE: 0.9 V (LPLV-STOP2 MODE), 1.25 V (RUN MODE)
Switch 1V8/3V3 for uSD
U205
NCP380HMU05AATBG
3-02772
U204
LP3990MF-1.8
3-01966
HSE
Y201
24 MHz
RESET
U203
STM1818RWX7F
3-02637
SW202
SW_DIP_x04
2-00008
LSE
Y203
32.768 kHz
6-00015
BOOT MODE
B2 B1 B0
0 0 0 UART/USB
0 1 1 NAND FLASH
1 0 0 TEST BOOT
1 1 0 UART/USB
/MPU Power & Clock/
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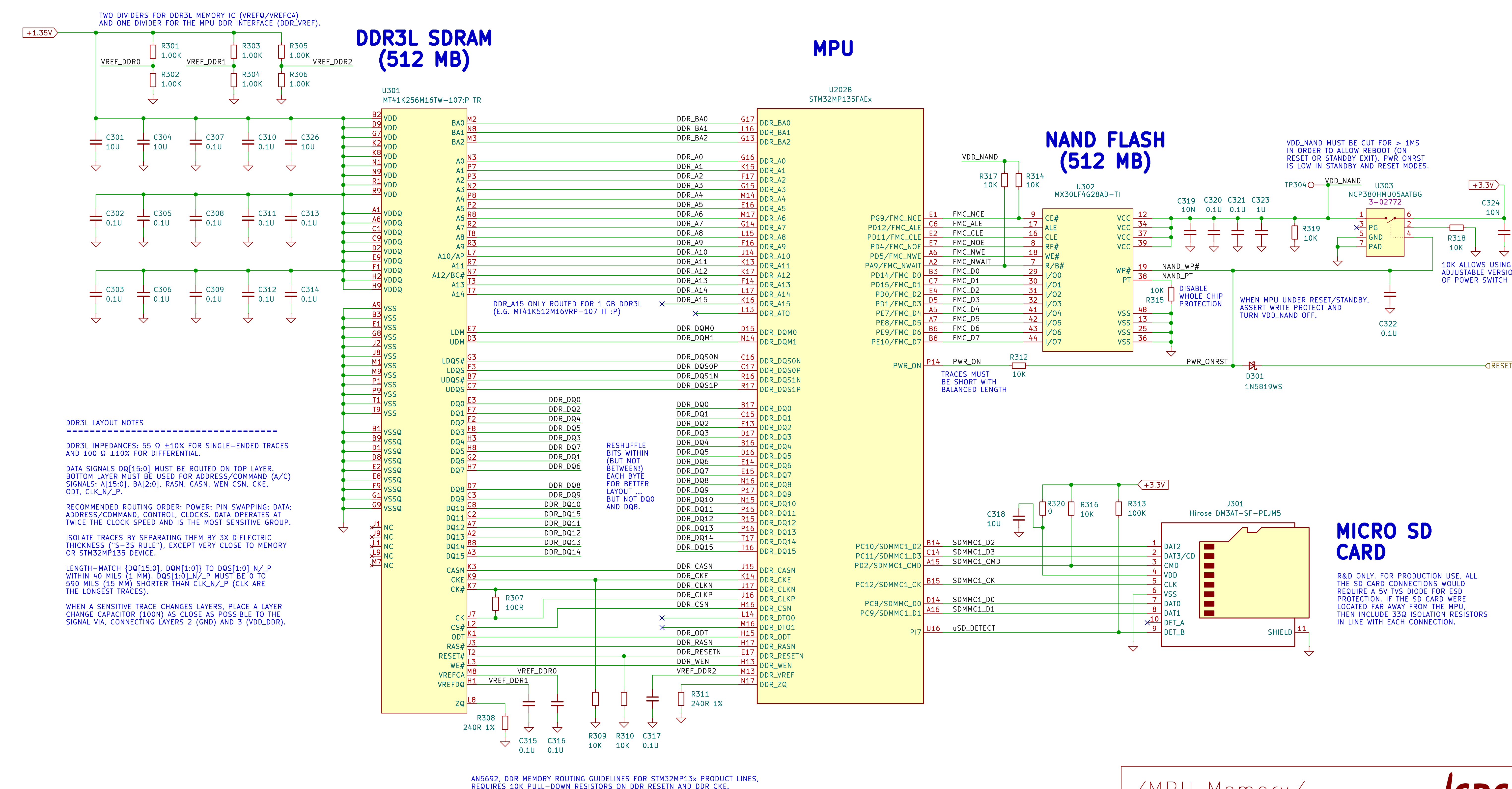
Sheet 2



/MPU Memory/

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STM32MP135 Test

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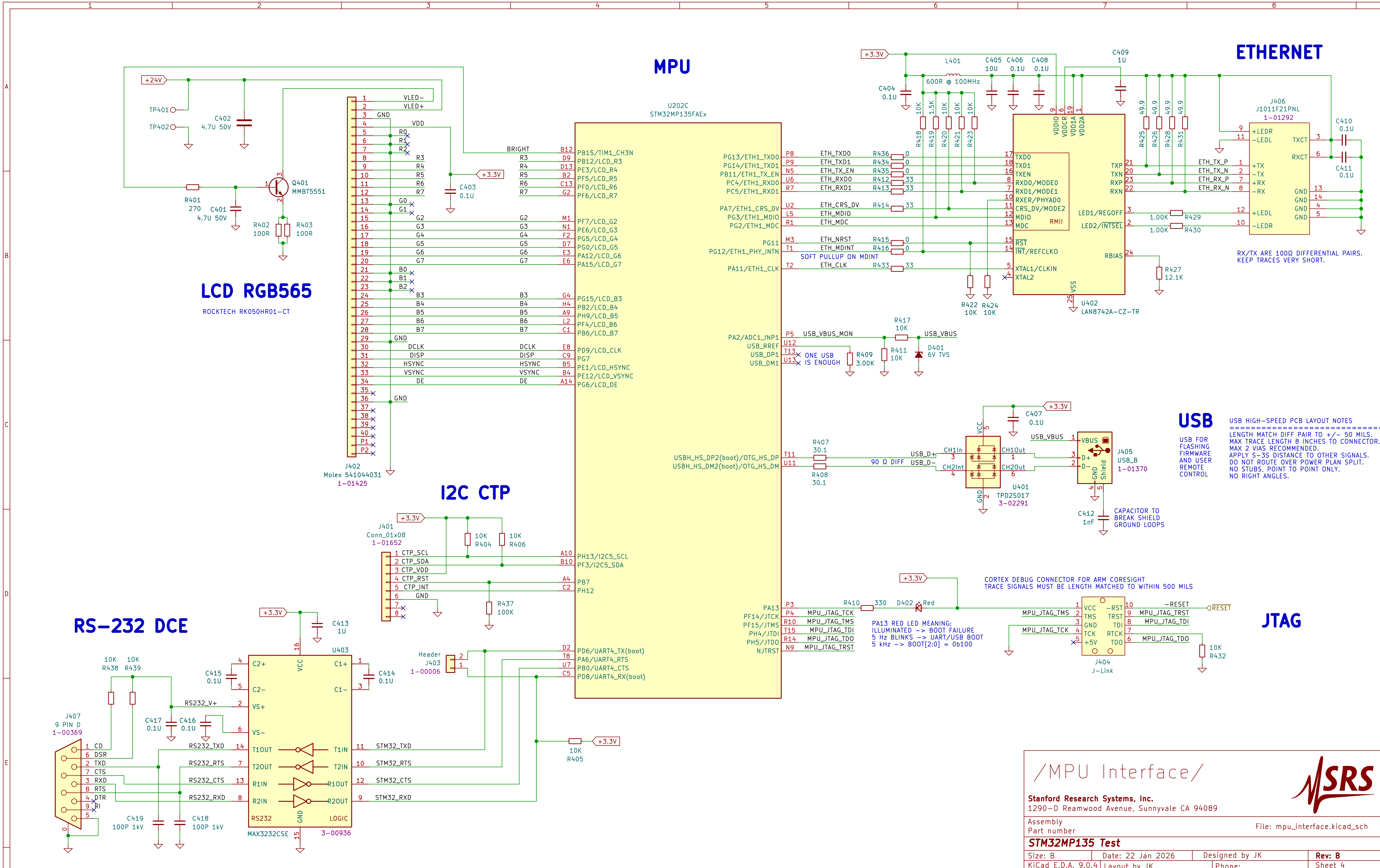

/MPU Interface/

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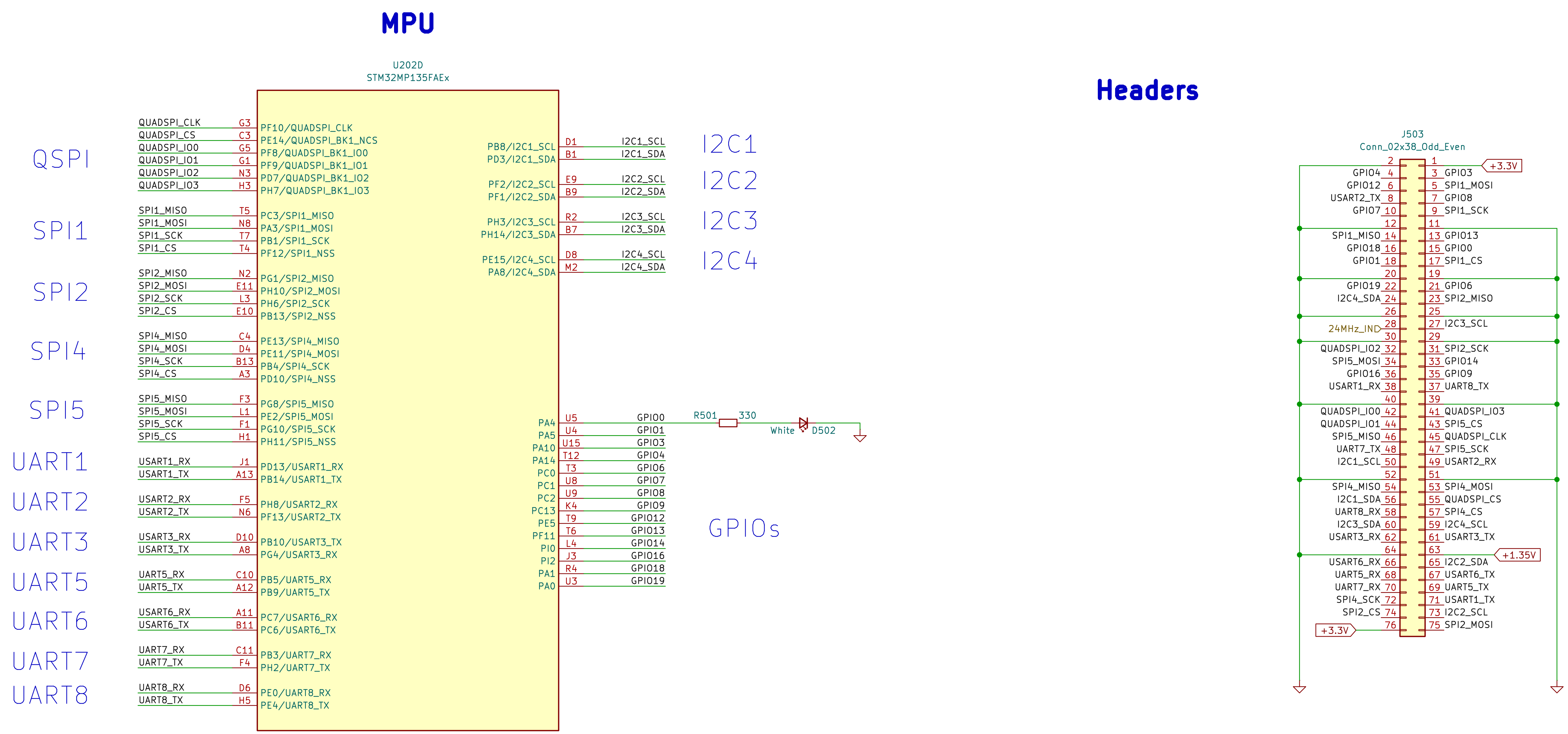

/MPU Extra Pins/

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STM32MP135 Test

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